

Progress Tracker — ML-Driven Inverse Design for Refrigerant Separation

A checkpoint counts as passed only when you can do it with AI and notes closed. A blank box is a precise map of what to restudy — not a reason to move on. Tick the box and date it the day you pass.

✓	CLOSED-BOOK CHECKPOINT (pass with AI + notes shut)	DATE
☐	0.1 Reproducible skeleton — Rebuild env from lockfile; dummy run reproduces.	
☐	0.2 Reusable math — Blank-file: closed-form Gaussian KL; state log-derivative identity.	
☐	0.3 Training loop — Blank-file: full PyTorch train loop with early stopping.	
☐	1.1 R-32/R-125 + IAST — Whiteboard azeotrope + how IAST gets mixture selectivity.	
☐	1.2 GCMC ground truth — Write GCMC acceptance rule; justify fugacity over pressure.	
☐	1.3 FF vs MLIP — Two-sentence FF defense from the MLIP comparison.	
☐	1.4 Featurize / no-leak — Quantify random-vs-grouped split gap; explain the leak.	
☐	2.1 PCA / reduction — Explain top components physically; why max-variance dirs.	
☐	2.2 Baseline + metrics — Argue when Spearman beats MAE for screening.	
☐	2.3 GNN message passing — Blank-file one MP layer; why sum is permutation-invariant.	
☐	2.4 UQ + calibration — Blank-file split conformal + reliability diagram; epistemic vs aleatoric.	
☐	3.1 Linker representation — Report random-validity rate; why naive sampling fails.	
☐	3.2 VAE from ELBO — Derive ELBO; why reparameterization is needed for backprop.	
☐	3.3 Diffusion from scratch — Blank-file 2-D DDPM fwd+rev; noise-prediction ~ score.	
☐	3.4 Conditioned generation — Plot the conditioning shift; explain guidance.	
☐	4.1 RL fundamentals — Write REINFORCE update; why a baseline cuts variance not bias.	
☐	4.2 Reward design — Defend every reward term; name one way it is gamed.	
☐	4.3 Steer the generator — Exhibit a reward-hack you caught + the guardrail.	
☐	5.1 Evaluation card — Explain valid+novel yet unsynthesizable; how card exposes it.	
☐	5.2 Active learning — Blank-file uncertainty sampling; how epistemic drives acquisition.	
☐	5.3 Wire the loop — Show before/after true-selectivity gain; which part drove it.	
☐	6.1 Reproducible pipeline — Collaborator reproduces one iteration from README alone.	
☐	6.2 Agentic layer — Agent completes a loop; you approve, never code the step.	
☐	6.3 Portfolio & JD map — For 3 postings, point to the artifact satisfying each bullet.	

Module payoffs: Module 2 → Paper 1 (calibrated selectivity screen) · Module 5 → Paper 2 (generative inverse design with simulation as the lab).
The honest test for every box: could you rebuild its core from a blank file and whiteboard it to someone?